



TECHNICAL DATA SHEET

ProLine PET-CF Filament

Product Overview

ProLine PET-CF is a modified PET-based 3D printing filament with 10% carbon-fiber reinforcement. It combines high temperature resistance, low shrinkage / low warping and easy printability, and can be processed on FDM printers without a heated chamber. Printed parts offer high rigidity and tensile strength and are suitable for long-term use at temperatures of up to 120°C.

Key Features

- Rigid, high-strength parts
- Low shrinkage / low warping for dimensionally stable prints
- Performs reliably at elevated temperatures (continuous use up to 120°C)
- Easy to print on non-heated chamber FDM printers

Material Properties

Physical Properties			
Property	Test Method	Unit	Value
Density	ISO 1183	g/cm ³	1.30–1.31
MFR (190°C / 2.16 kg)	ISO 1133	g/10 min	3–6
Moisture Absorption (23°C / 24 h)	ISO 62	%	< 0.3
Mechanical Properties			
Tensile strength (X-Y)	ISO 527	MPa %	75–85 4–5
Elongation at break (X-Y)	ISO 527	% MPa	1.5–2.5
Elongation at break (X-Z)	ISO 527	MPa	4500–5000
Flexural modulus (X-Y)	ISO 527	MPa	2000–2500
Flexural modulus (X-Z)	ISO 527	kJ/m ²	140–150 25–
Flexural strength (X-Y)	ISO 178		30
Impact strength (X-Y)	ISO 180		
Thermodynamic Properties			
HDT @ 0.455 MPa (66 psi)	ISO 75	°C	190
Continuous Use Temperature	IEC 60216	°C	120

Recommended Printing Parameters

Parameter	Setting
Print Temperature	280-300°C
Bed Temperature	80-100°C
Bed Materials	Tempered glass, BuildTak, Carbon fiber board
Nozzle Diameter	Ø 0.4 / 0.6 mm
Nozzle and gear material	High strength steel
Model Cooling Fan	OFF

Drying & Handling Notes

- Pre-dry: 90-100°C, minimum 12 h (hot-air oven) to reduce moisture-related printing issues and improve surface/part quality
- Post-anneal: 80-100°C, 8 h to increase part strength

Disclaimer

Product information and test results in this document were obtained under controlled laboratory conditions. Real-world performance may vary depending on processing, environment, and application. The customer is responsible for determining product suitability for the intended use and for ensuring compliance with all applicable laws and regulations. The Company accepts no liability for the use of this information and provides no warranties, including any implied warranties of merchantability or fitness for a particular purpose, to the extent permitted by law.

Test Specimens
Test sample dimensions and printing direction

Figure 1. Tensile testing specimen; ASTM D638 (ISO 527, GB/T 1040)

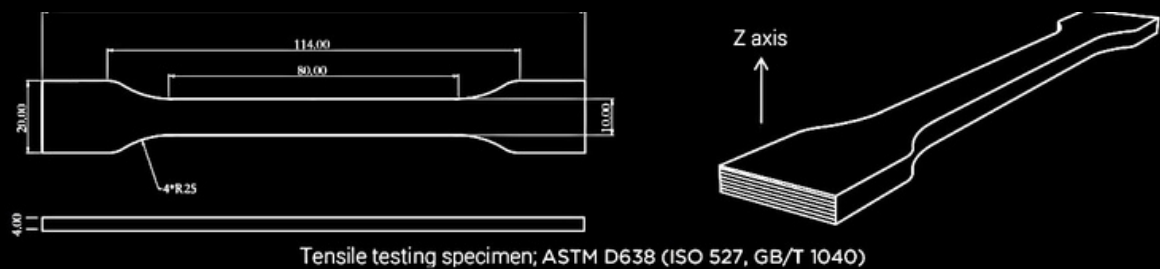


Figure 2. Flexural and impact testing specimens; ASTM D790 (ISO 178, GB/T 9341) and ASTM D256 (ISO 179, GB/T 1043)

